

Sustainable Development

Green Building Energy Management Portal

Individual Technical Report

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ABSTRACT

Buildings consume a large amount of electricity for lighting, air conditioning, computers, appliances, and other systems. Unnecessary energy consumption increases electricity costs and contributes to environmental pollution and greenhouse-gas emissions.

The proposed Green Building Energy Management Portal is a responsive web-based system that helps users monitor energy consumption, calculate electricity costs and carbon emissions, and identify areas of energy wastage. The main individual contribution is an AI-assisted recommendation system that provides personalized energy-saving suggestions based on consumption patterns.

The system can be developed using HTML, CSS, and JavaScript. It supports sustainable energy use and contributes to SDG 7, SDG 11, SDG 12, and SDG 13.

INTRODUCTION

Energy efficiency is an important part of sustainable building management. Many buildings waste electricity because lights, air conditioners, computers, and other equipment are used inefficiently.

Traditional electricity bills only show overall consumption and cost. They do not clearly indicate which appliances consume the most energy or how users can reduce consumption.

The proposed portal uses Information Technology to make energy consumption easier to monitor and understand. It provides energy data, calculations, graphs, and recommendations through a simple responsive website.

PROBLEM STATEMENT

Buildings often experience high electricity consumption due to inefficient use of electrical appliances and lack of proper monitoring.

Problem:

How can an IT-based system monitor building energy consumption, identify energy wastage, estimate environmental impact, and suggest ways to improve energy efficiency?

SUSTAINABLE ANALYSIS

- Sustainable Development Goals

SDG 11 – Sustainable Cities and Communities

SDG 12 – Responsible Consumption and Production

SDG 13 – Climate Action

- ISO 14001

ISO 14001 is an international standard for Environmental Management Systems (EMS). It helps organizations reduce environmental impacts and improve their environmental performance.

The proposed portal supports ISO 14001 by:

- Monitoring energy consumption.
- Identifying energy wastage.
- Tracking environmental performance.
- Supporting energy-reduction targets.
- Promoting continuous improvement.

- ## Life Cycle Assessment

Life Cycle Assessment (LCA) evaluates environmental impacts throughout the life cycle of a product or system.

The proposed system mainly focuses on the operational stage of a building, where electricity is consumed for lighting, cooling, appliances, and other activities.

Reducing operational energy consumption can help reduce the environmental impact of building operation.

PROPOSED ENGINEERING/ IT SOLUTION

My solution is an AI-Assisted Green Building Energy Management Portal.

The system consists of:

- Energy Dashboard
- Energy Monitoring
- Energy Calculator
- Carbon Emission Calculator
- Energy Efficiency Score
- Green Recommendations

Users can enter appliance details such as power rating and operating hours. The system calculates energy consumption and provides suitable recommendations.

The basic system can use manually entered data. In future, smart meters and IoT sensors can be used for automatic monitoring.

FEASIBILITY ANALYSIS

Technical Feasibility

The system can be developed using HTML, CSS, and JavaScript, which are widely available and suitable for a student project.

Economic Feasibility

The basic website requires minimal development cost. Future IoT devices and server infrastructure may require additional investment.

Operational Feasibility

The portal has a simple interface and can be used by students, employees, building administrators, and other users.

Environmental Feasibility

The system encourages reduced energy consumption and can therefore help reduce associated environmental impacts.

EXPECTED ENVIRONMENT BENEFITS



The proposed system can:

- Reduce unnecessary electricity consumption.
- Reduce associated carbon emissions.
- Improve energy awareness.
- Encourage energy-efficient appliances.
- Promote renewable energy.
- Support better building management.
- Encourage sustainable user behaviour.



JUSTIFICATION OF THE PROPOSED SOLUTION

The proposed solution is useful because it converts energy data into understandable information and actionable recommendations.

Unlike a normal electricity bill, the portal allows users to see consumption patterns and identify areas of wastage. The personalized recommendation feature makes the system more practical because users receive suggestions based on their energy usage.

The system is also scalable. It can begin with manual data entry and later be connected to smart meters, IoT sensors, and machine-learning systems.

Therefore, the solution is technically feasible, economically practical, environmentally beneficial, and suitable for green building management.

CONCLUSION

The Green Building Energy Management Portal provides an IT-based solution for improving energy efficiency in buildings. It combines energy monitoring, calculations, visualization, carbon-emission estimation, and personalized recommendations.

The main individual contribution is the AI-assisted energy recommendation system, which helps users understand their energy consumption and take appropriate actions to reduce wastage.

The system supports SDG 7, SDG 11, SDG 12, and SDG 13 and can contribute to sustainable building management. With future integration of IoT and machine learning, the portal can become a more advanced and intelligent energy-management system.

REFERENCES

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